Applied Optimization - Ex 3

QCQP to SOCP

1.) ($\Leftarrow$) $\left\| \begin{matrix} \frac{1+(b^Tx+c)}{2} \\ Ax \end{matrix} \right\|_2 \leq \frac{1-(b^Tx+c)}{2}$ (let $u = b^Tx+c$, and square both sides)

(Note: $\left\|\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}\right\|_2^2 = \left\|\begin{bmatrix} x_2 \\ \vdots \\ x_n \end{bmatrix}\right\|_2^2 + x_1^2$)

$$\|Ax\|_2^2 + \left(\frac{1+u}{2}\right)^2 \leq \left(\frac{1-u}{2}\right)^2$$

$$\|Ax\|_2^2 + \frac{1+2u+u^2}{4} \leq \frac{1-2u+u^2}{4}$$

$$\|Ax\|_2^2 + \frac{4u}{4} \leq 0$$

$$x^TA^TAx + b^Tx + c \leq 0$$

($\Rightarrow$) $x^TA^TAx + b^Tx + c \leq 0$

$$\|Ax\|_2^2 + u \leq 0$$

$\vdots$

Reverse algebra above

$$\left\| \begin{matrix} \frac{1+(b^Tx+c)}{2} \\ Ax \end{matrix} \right\|_2 \leq \frac{1-(b^Tx+c)}{2} \quad \blacksquare$$

2.) minimize $x_1^2 + 4x_1x_2 + 4x_2^2 = (x_1+2x_2)^2$

s.t. $9x_1^2 + 16x_2^2 \leq 25$ } already in SOCP form
$x_1 - x_2 = 1$

$\Rightarrow$ epigraph transform

minimize$_{x,t}$ t
s.t. $x_1^2 + 4x_1x_2 + 4x_2^2 \leq t$ (quadratic inequality constraint)

$\Rightarrow$ algebra

$$x_1^2 + 4x_1x_2 + 4x_2^2 = x^THx = \underbrace{x^TA^TAx}_{\text{for some } A} = \|Ax\|_2^2$$

$\Rightarrow$ $\|Ax\|_2 \leq \underbrace{\sqrt{t}}_{\text{not affine}}$

$\Rightarrow$: $\|Ax\|_2 \leq s$, $\underbrace{s^2 \leq t}_{\text{Not in SOCP form}}$

$\Rightarrow$ $\|Ax\|_2 \leq s$, $(2s, 1, t) \in \underbrace{Q_r^3}_{\text{rotated second order cone}}$

Final problem

minimize$_{x,t}$ t
s.t. $\left\|\begin{bmatrix} 3x_1 \\ 4x_2 \end{bmatrix}\right\|_2 \leq 5$
$x_1 - x_2 = 1$
$\|Ax\|_2 \leq s$
$(2s, 1, t) \in Q_r^3$

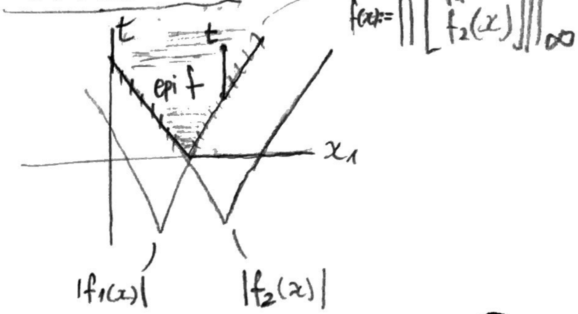
Linear Programming

$$\text{minimize} \quad \|(2x_1 + 3x_2, -3x_1)^T\|_\infty$$
$$\text{s.t.} \quad |x_1 - 2x_2| \le 3$$

(1) Note $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$

$$\begin{aligned} &\underset{x,t}{\text{minimize}} \quad t \\ &\text{s.t.} \quad |2x_1 + 3x_2| \le t \\ &\qquad |-3x_1| \le t \\ &\qquad x_1 - 2x_2 \le 3 \\ &\qquad -(x_1 - 2x_2) \le 3 \end{aligned} \Bigg\} \text{ epigraph transform.}$$

Visualize : $x \in \mathbb{R}^1$



$$\begin{aligned} &\underset{x,t}{\text{minimize}} \quad t \\ &\text{st} \quad \pm(2x_1 + 3x_2) \le t \\ &\qquad \pm(-3x_1) \le t \\ &\qquad \pm(x_1 - 2x_2) \le 3 \end{aligned} \Bigg\} \text{ 6 constraints total (2 for each } \pm)$$

(2) (a) All vars in $\mathbb{R}_+$

$$\begin{aligned} &\underset{x,t}{\text{minimize}} \quad t \\ &\text{s.t.} \pm(2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-)) \le t \\ &\quad \pm(-3(x_1^+ - x_1^-)) \le t \\ &\quad \pm((x_1^+ - x_1^-) - 2(x_2^+ - x_2^-)) \le 3 \\ &\quad x_1^+, x_1^-, x_2^+, x_2^- \ge 0 \end{aligned}$$

(b) Remove general inequality constraints

$$\begin{aligned} &\underset{x,t}{\text{minimize}} \quad t \\ &\text{s.t.} \quad 2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-) - t + s_1 = 0 \\ &\qquad -(\quad '' \quad) - t + s_2 = 0 \\ &\qquad -3(x_1^+ - x_1^-) - t + s_3 = 0 \\ &\qquad 3(x_1^+ - x_1^-) - t + s_4 = 0 \\ &\qquad (x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) - 3 + s_5 = 0 \\ &\qquad -(\quad '' \quad) - 3 + s_6 = 0 \\ &\qquad x_1^+, x_1^-, x_2^+, x_2^-, s_1, \dots, s_6 \ge 0 \end{aligned}$$

Also,
t = t^+ - t^-
t^+, t^- >=0
for all t above.